

Assignment: Task 1 – APT Attack Simulation and Analysis

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Course: DLBCSEECTI02_E – Defense Against APTs

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1. Introduction

Advanced Persistent Threats are one of the most pressing issues in contemporary cybersecurity. Unlike common cyber-attacks, APTs are characterized by their stealthiness, targeted, and prolonged nature. Their primary objective is to penetrate a system, remain hidden for an extended period, and extract sensitive information. Such threats pose a severe threat to corporations, particularly those dealing with confidential information or critical infrastructure.

This research project aims to establish a controlled setting where an attack simulation using Advanced Persistent Threat (APT) methods will be possible. This will allow for the evaluation of existing security systems and identify their vulnerabilities. Through this research, the behavior of the attacker will be understood and used to develop more efficient defense strategies against the particular attack methods.

In this research, the cyber criminal organisation APT29, known also as Cosy Bear, was employed as an example. An attack scenario was designed and executed based on the strategies and tactics associated with APT29 as provided by the MITRE ATT&CK® framework in a simulated environment (MITRE ATT&CK, 2026). The simulation focuses mainly on the key stages such as initial access, persistence, and C2 communication. In parallel to this process, logs from the systems as well as other information were collected to perform the analysis. The data collected in this way allowed conducting a gap analysis to detect any gaps in detection and response systems.

2. Main Body

2.1 APT29 and Attack Scenario Design

APT29, also referred to as Cosy Bear, is one of the many Advanced Persistent Threats selected as an example to be examined in this assignment. Specifically, APT29 is an espionage group of Russian Foreign Intelligence Service, which normally focuses its attacks on governmental agencies, scientific organizations, and corporations within NATO countries (MITRE ATT&CK, 2025). As a rule, it maintains its presence on compromised networks for quite a long period while avoiding detection. Therefore, such an attack could be used as a good example of testing the abilities of the defenders. APT29 applies tactics related to stealthiness and persistence and normally uses legitimate tools of the operating system under

attack. Such tactic, known as "Living off the Land," allows attackers to blend malicious activities into the operation of a system and bypasses usual protective procedures.

The attack scenario for this project is based on APT29 tactics and techniques that have been documented in the MITRE ATT&CK® framework. The scenario starts with a fake phishing attack that gives the attacker access to the system and makes the user unknowingly run a payload. Phishing can be done through emails. After getting in, the attacker uses a scheduled task to make sure that they can stay in the system and keep running.

Command and control (C2) communication is set up to interact remotely with the compromised system (MITRE ATT&CK, 2025). During this stage, the attacker runs commands and interacts with the system while trying to stay hidden.

2.2 Simulation Setup and Execution

A controlled virtual environment was set up to safely copy the interactions between the attacker and the target in order to simulate the attack scenario. Virtualisation made it possible to run the simulation without affecting real systems, while still providing a useful and observable setup. There were two virtual machines set up in a local virtual network that made up the environment. The target system was a Windows 11 computer, which is what most people would use. The attacker used a Kali Linux machine to carry out and control the attack activities.

The attacking system as well as the victim could communicate via a private internal network that was built between the two VMs. In this experiment, I performed some basic network configurations such as checking IP addresses as well as network communication between systems.

Before the experiment took place, the firewall that came installed within the windows was turned off so that the communication would happen without any restrictions. It is crucial to emphasize the fact that this setup does not reflect a safe real-world environment, since firewalls are crucial for the security of a system.

Also, logging features were turned on on the target system so that it could record system activity. This includes Windows Security Event Logs, which keeps track of things like logon activity and process execution. This setup makes it possible to collect data, which was then used to look at how the attacks worked.

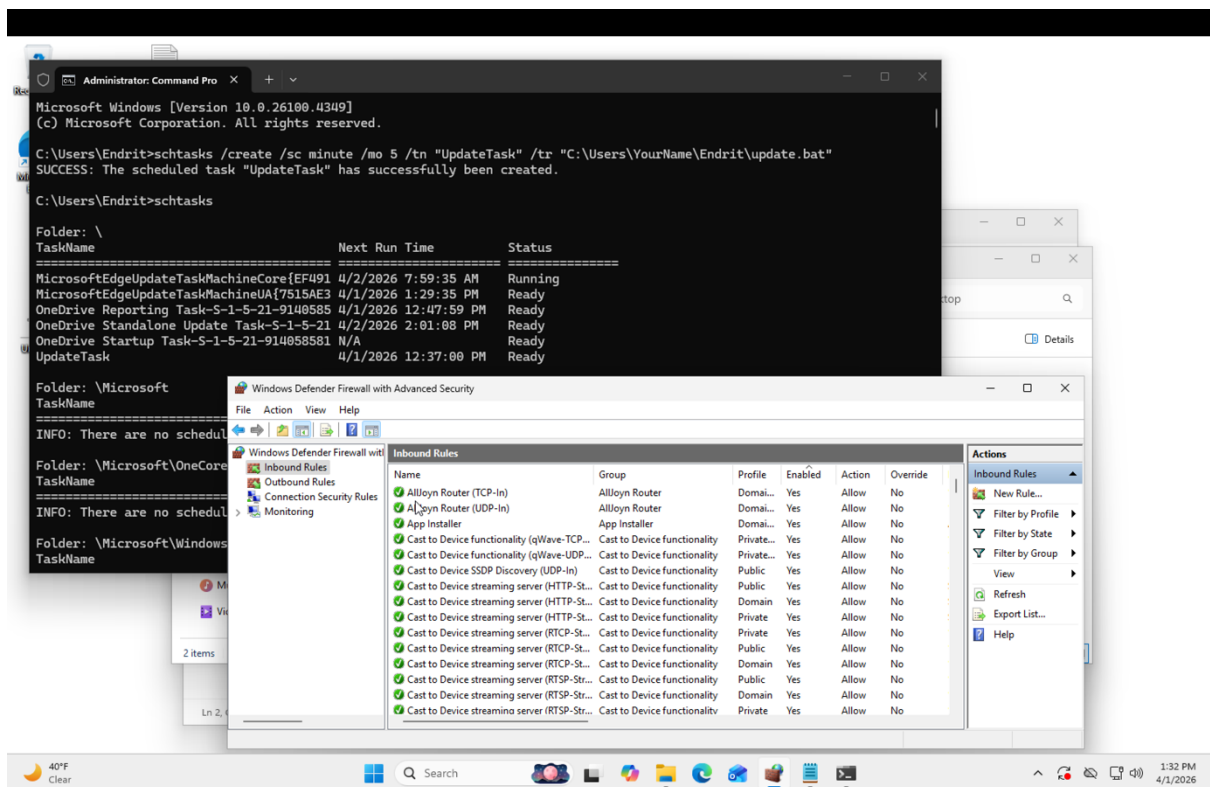


Figure 1: Enabling all Inbound Rules in the Windows Firewall settings.

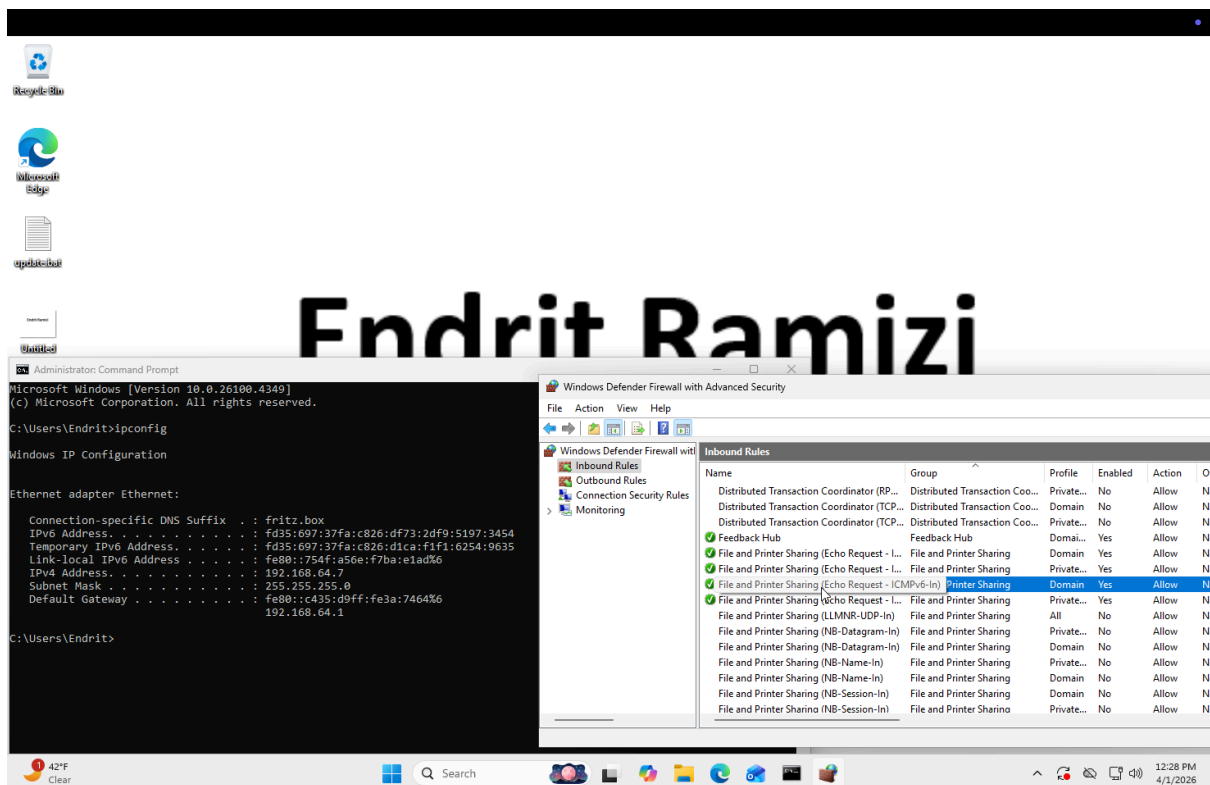


Figure 2: Network configuration showing the IP address of the target system.

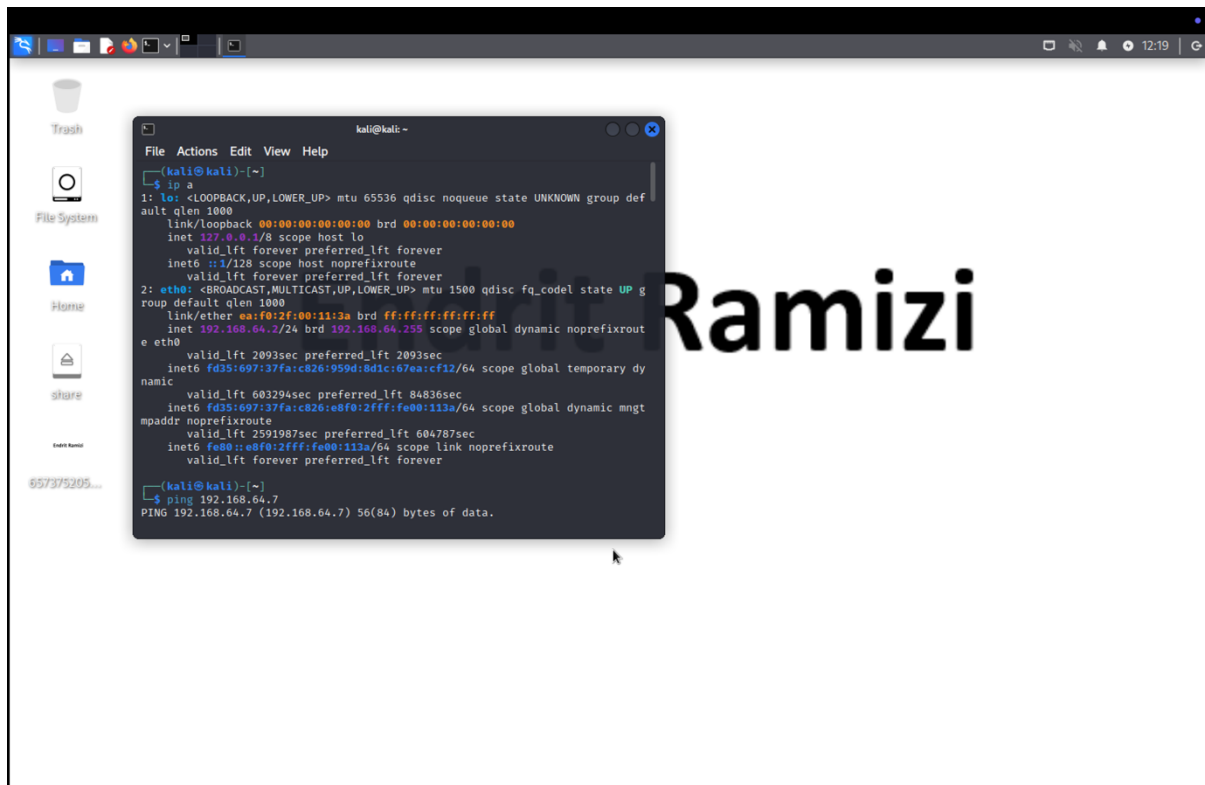


Figure 3: Network configuration showing the IP address of the attacker system.

2.3 Execution of the APT29 Attack Simulation

The attack simulation followed a sequence of steps according to the nature of advanced persistent threats (APTs) as outlined by MITRE ATT&CK. In this way, certain stages of attack progression could be simulated in a controllable environment for observations. The first stage was the one of establishing a connection between the attacking machine (Kali Linux) and the attacked system (Windows). This was done via network-based commands, such as ping, to detect the system and establish communication. At this stage, Nmap was used to detect the ports of the system that could be exploited later (Nmap, n.d.).

Prior to gaining access, Nmap was used to perform a simple network scan on the target system to detect open ports on it. The following open ports were identified, including 135, 139, and 445 that are commonly known for being used in Windows networking services (Microsoft, 2026). The port 135 is utilized for the Remote Procedure Call (RPC) that provides connectivity between services in a Windows operating system. The port 139 is involved in NetBIOS Session Service and used for file sharing purposes in Windows networking environment. Port 445 is associated with the use of Server Message Block (SMB) protocol used for file sharing as well.

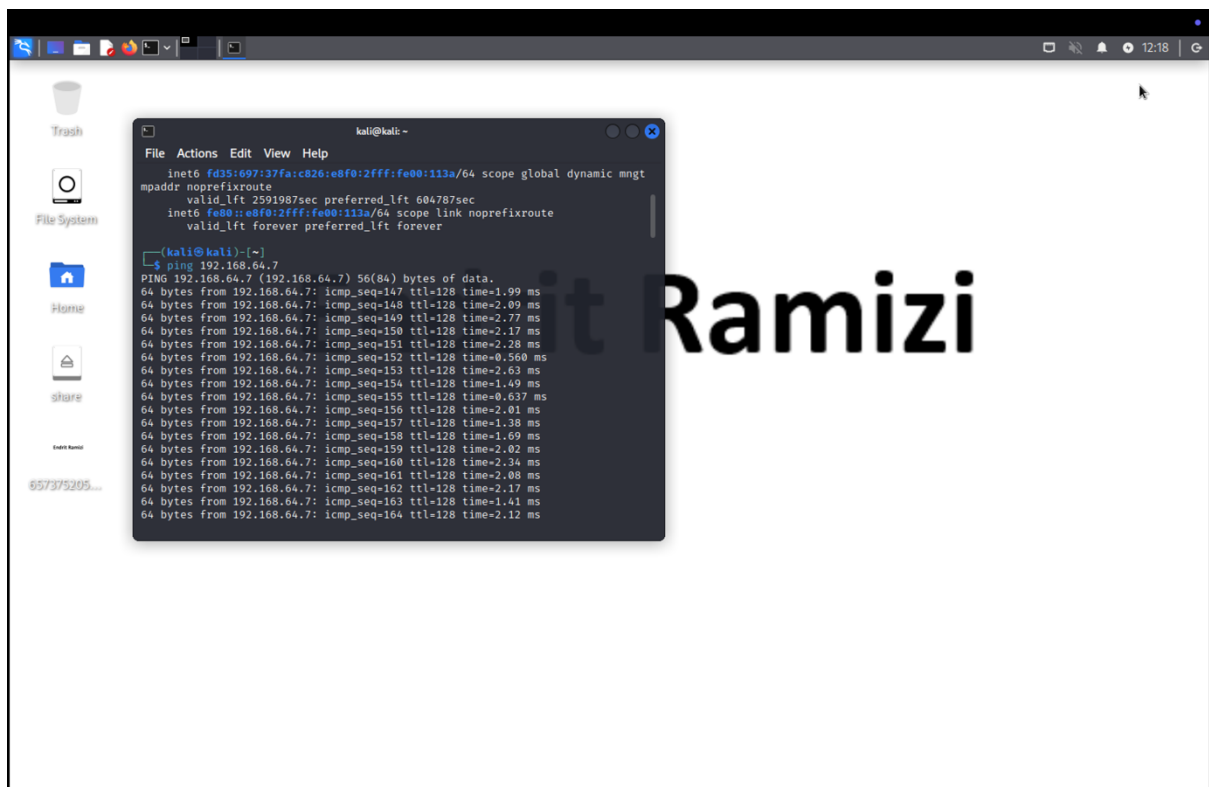


Figure 4: Connectivity between the attacking machine with the target machine. (Kali).

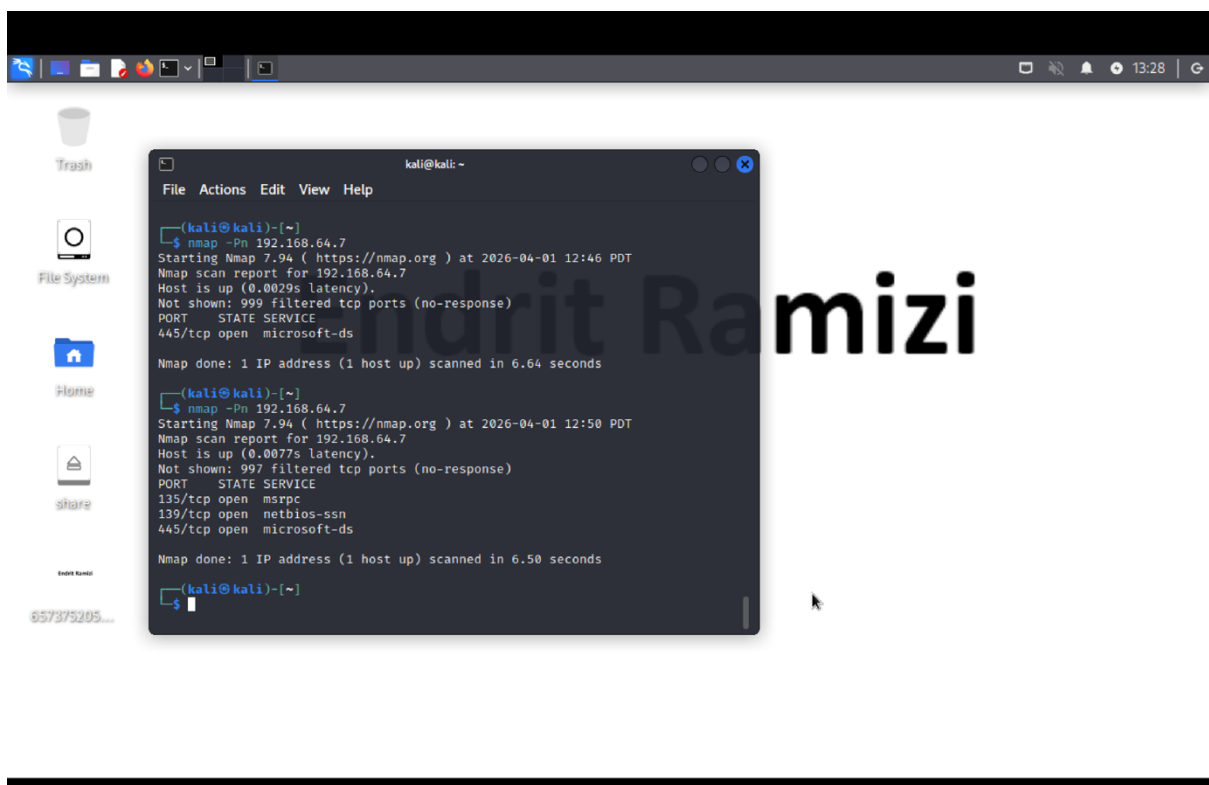


Figure 5: Scan showing multiple open ports (135, 139, 445).

After gaining entry, a task was created to ensure persistence. An update task was created using the command prompt. This task would be used to automatically execute a script (update.bat) every five minutes. As a result, it ensured that the attacker maintained persistence in the system by executing the script automatically without needing any input from the user. This technique resembles the persistence tactics used by sophisticated malware.

Following this, the two systems were connected remotely for the purposes of conducting C2 communication. The attacker is thus able to communicate with the target computer and issue commands, this can be described as remote control, a process commonly employed by attackers in the real world.

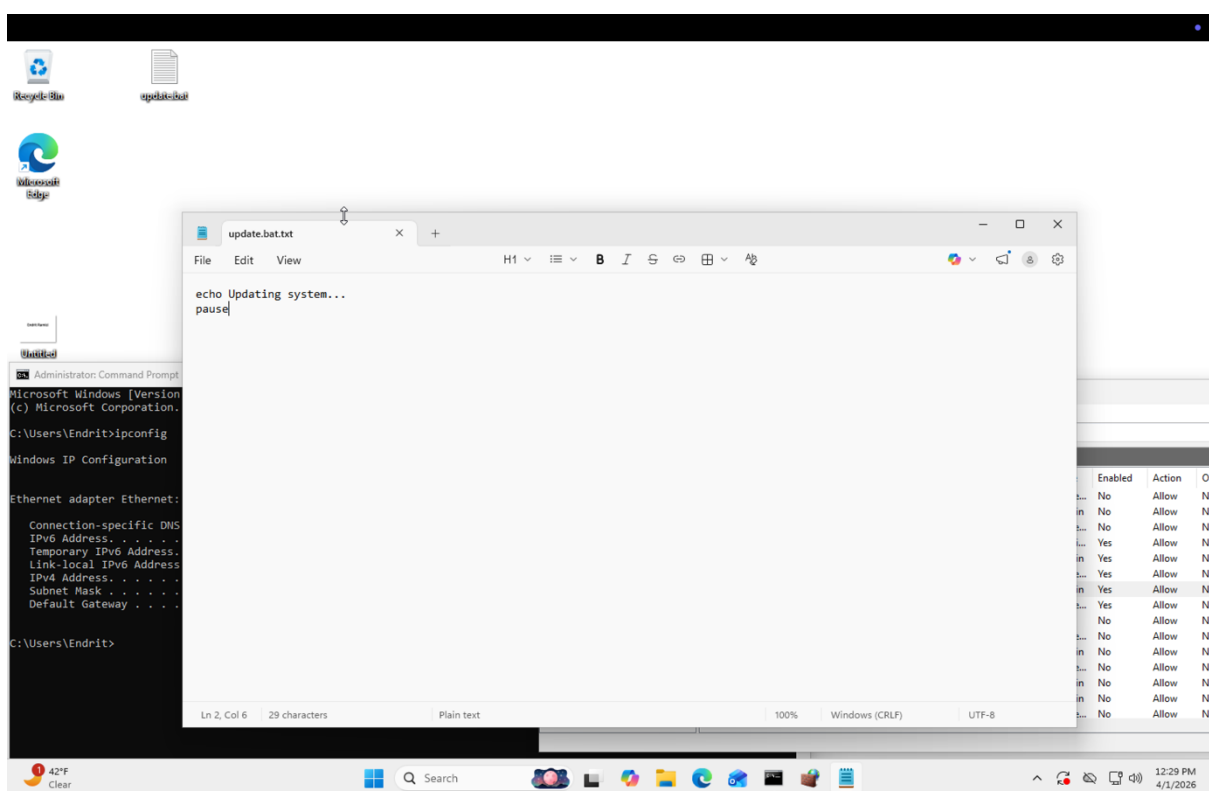


Figure 6: Malicious script used to simulate phishing-based initial access

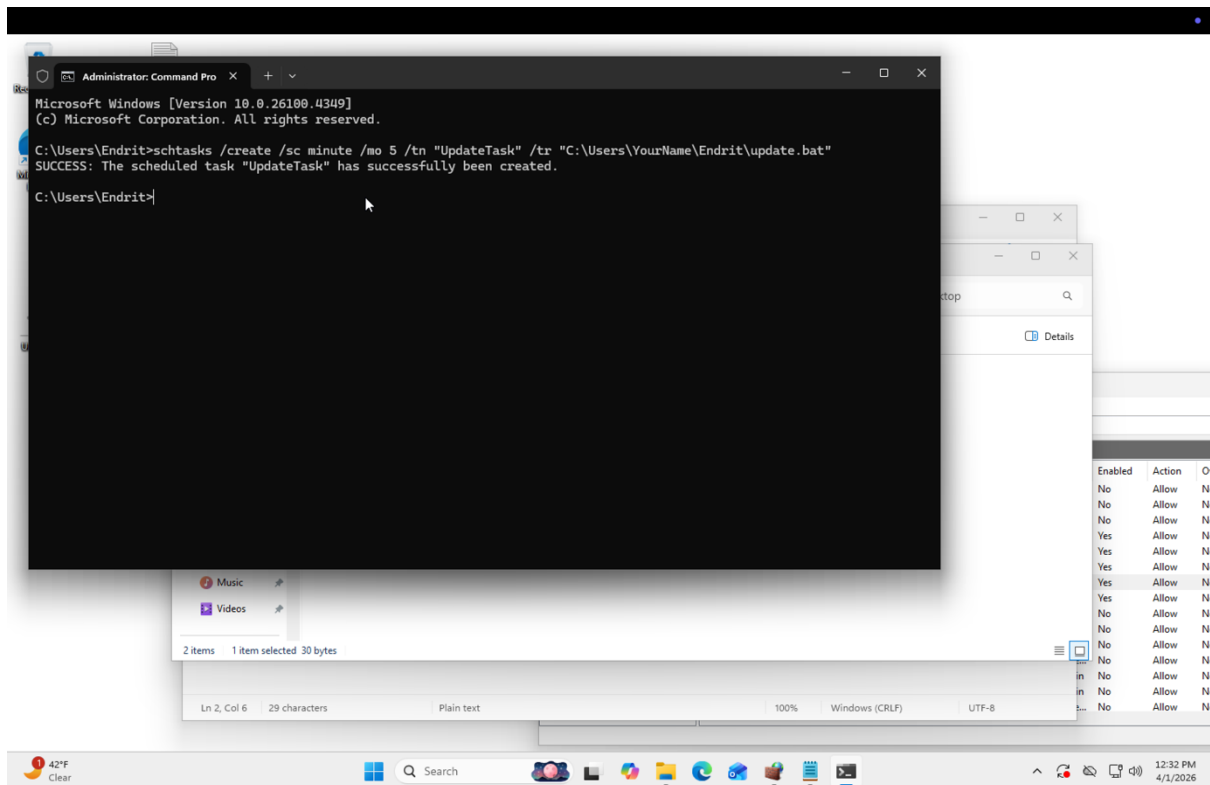


Figure 7: Creation of scheduled task for persistence

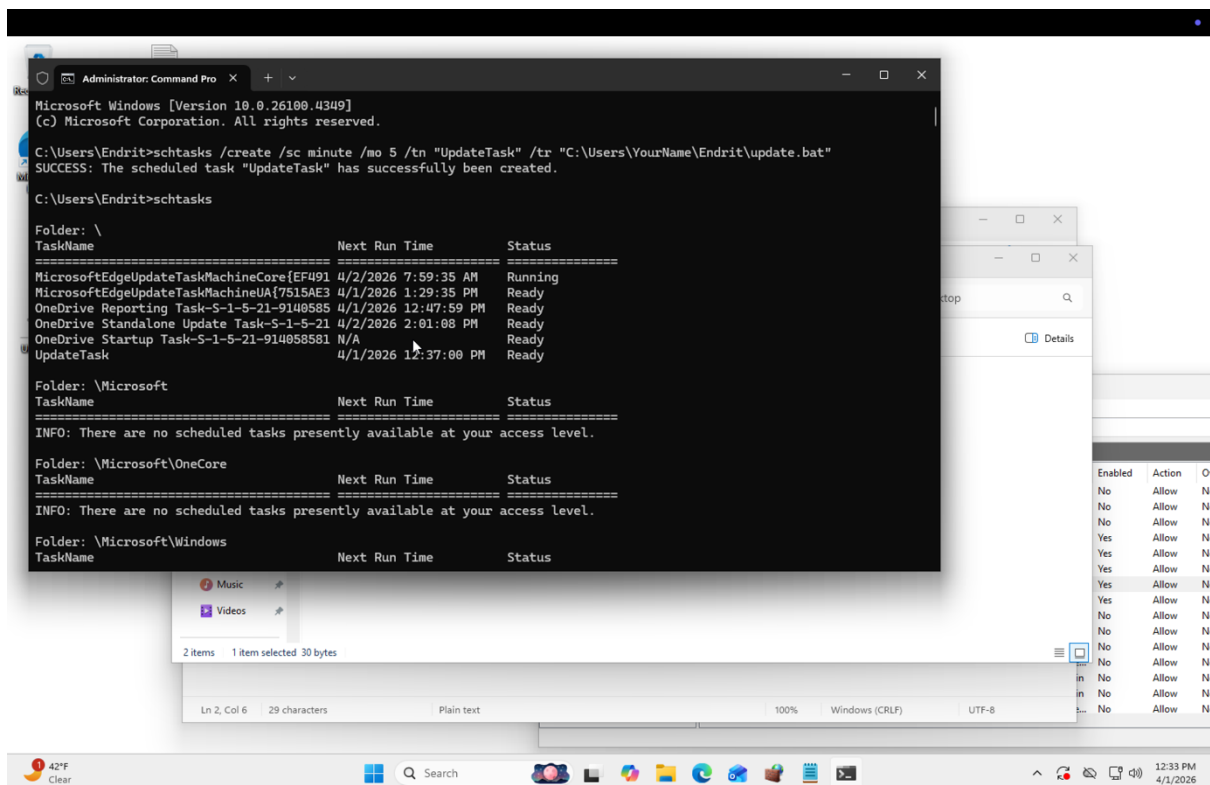


Figure 8: Verification of persistence through task listing



Figure 9: Command and Control (C2) communication established between attacker and target system.

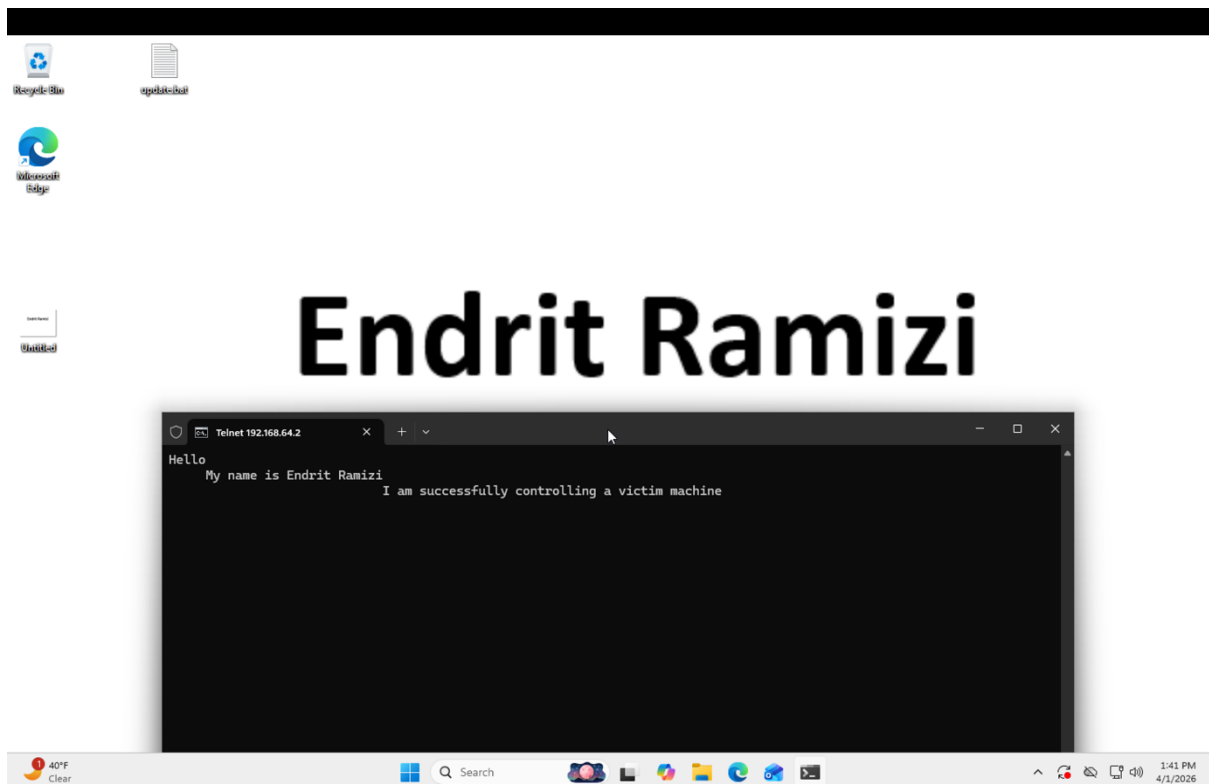


Figure 10: Command and Control (C2) communication result.

For this simulation, the activities on the system were generated, and then examined using Windows utilities that came with the system. Special attention was paid to the behavior of the system and the log files during each of the attacks. Although the simulation was relatively small-scale, it provided some interesting insights into remote access to a computer system and system interactions, and the difficulties involved in detecting any malicious behavior through system logs. For safety and consistency, everything was performed in a controlled virtual environment.

2.4 Data Collection and Analysis

Various types of data were collected during the attack simulation in order to examine the system's performance and to evaluate how effective the security protocols implemented by the latter had been. Windows Security Event Logs and Event Viewer provided the primary data sources, revealing system activity (Microsoft, 2026). During the simulated attack, system activity was monitored in order to document significant events, such as logons, process execution, and scheduled tasks. The event logs provided insights into the dynamics of the attack and its various stages.

The analysis showed that certain aspects associated with the attack had left traceable evidence. For instance, in the C2 phase, there were many instances of successful logon events (Event ID 4624), privileged logons (Event ID 4672) and process creation (Event ID 4688). Creation of scheduled tasks (UpdateTask) also acted as a form of persistence, which is known to always leave evidence in the system's logs. However, most of these activities did not raise much alarm from the normal system's activities. The logs consisted of distinct events without any indication of any malicious acts. Thus, it was impossible to determine that the activities were performed by an attacker unless there was a time reference point.

Furthermore, from the analysis, it could be seen that there was little association between incidents despite the availability of logs. There was no automatic association of events, making it difficult to detect such attacks. This highlights the significance of having centralized logging and correlation of events in the process of detecting complex attacks.

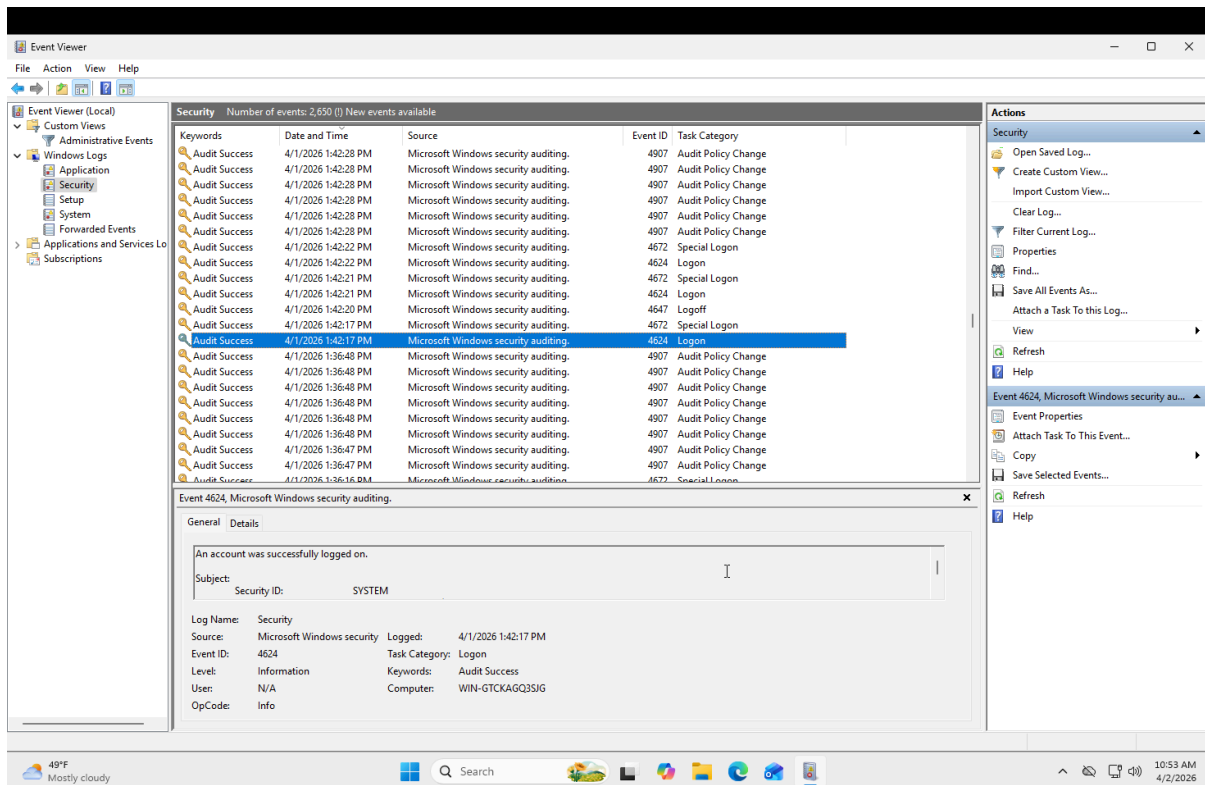


Figure 11: Security logs observed during the command and control phase using Windows Event Viewer.

2.5 Gap Analysis results

This gap analysis was done by analyzing the information collected through the simulation and comparing it against the effectiveness of the security systems in place to detect any threats in the environment. It specifically focused on determining whether there were activities during the attack simulation that were recorded and others that were not easily identifiable as threats. As indicated from the findings, although there were indicators generated for some attack activities, detection of such activities remained inadequate. An example is the generation of many indicators during the C2 and first access phases, such as logon and creation of processes. However, they were indicative of normal operations rather than threats (MITRE Corporation, 2021).

There was also a significant flaw identified with regards to persistence techniques. Despite being able to establish UpdateTask, which was a scheduled task to ensure that the script was always executing, this did not trigger any alarms or warnings. Hence, it is evident that monitoring for task scheduling activities is not sufficient, and attackers can leverage this to maintain persistent access to the victim machine. Logs were created for C2 activities, although they resembled normal system events and were difficult to identify as malicious

actions. In other words, the simulated cyberattack was able to hide within legitimate activities since it made use of standard system utilities and normal network traffic.

Another major issue that arose from this investigation was the inability to correlate events (MISP Project, 2025). Logs were generated in the process of the simulation, however, these logs were then analyzed separately, not collectively. Therefore, it was impossible to determine any connections between these events and recognize them as a coordinated cyberattack. As shown by gap analysis, no advanced methods for detecting threats were incorporated into the simulated environment. Specifically, there were no alerting mechanisms combined with centralized log analysis. Such vulnerabilities enabled the cyberattack to take place undetected, thus demonstrating the importance of improving detection mechanisms in cybersecurity environments.

2.6 Defense Recommendations

Based on the results of the gap analysis, there are several potential measures that may be undertaken in order to enhance the overall level of security within the system and improve detection and response capabilities to threats. On a technical basis, it is essential to implement an advanced monitoring and detection system. In order to facilitate centralized logging and correlate security incidents, SIEM systems need to be implemented (National Institute of Standards and Technology, 2012). This will help identify certain trends such as repetitive logon attempts and creation of new processes that occurred during the simulation exercise but were not recognized as malicious actions.

Moreover, it is recommended that you use EDR solutions for monitoring endpoints and detecting any unusual activities. For example, creating and executing scheduled tasks such as the UpdateTask used for persistence during this simulation must trigger alerts or require further analysis. To ensure visibility, there must be improvements made in the audit process. Specifically, comprehensive logging for scheduled tasks, authentication events, and processes must be enabled. Instead of reviewing logs after the incident, you should proactively monitor them.

Also, anomaly-based detection solutions must be considered in order to identify any change in system operation patterns. Behavioral analysis can aid in detecting any small indicators of abnormal operation which may go unnoticed by signature-based detection techniques due to simulated C2 traffic being injected into system operations. Apart from technical measures, organizational changes are equally necessary. In order to reduce the chance of successful user-based attacks such as executing an untrusted file, regular

employee training on safe practices must be provided. In addition, a proper security policy must be established for efficient incident response.

Last, but not least, the company's security policy must incorporate regular security assessments through techniques such as penetration testing and attack simulation. Such measures will help identify vulnerabilities. In order to be able to discover and prevent attacks similar to the above-mentioned one, several actions have to be taken.

3. Conclusion

The objective of this project was to simulate an attack using Advanced Persistent Threat (APT) methodologies, taking APT29 as an example. This experiment was aimed at assessing the level of security of the system through the eyes of an attacker. Through this experiment, many insights can be gained on how an attacker behaves and what difficulties there are in detecting this kind of attack. The result showed that some attacks leave clear marks in the system log file, but others do not have any clear signs of being malicious.

The gap analysis reveals that the monitoring, event correlation, and event detection capabilities were very weak. Although logs existed, they could not be correlated as part of an attack that had been executed in a sequence of events since there was no central system of analyzing and alerting. All these factors highlight the need to consider better methods like log correlation and behavioral detection as well as advanced monitoring capabilities. The proposed defense strategies address these issues using technological and organizational solutions.

In summary, it can be noted that the project proves the effectiveness of attack simulation in terms of detecting vulnerabilities and enhancing defense strategies. The hands-on learning experience provided me with valuable insights into the ways attacks may appear within the logs of a system and the attempts made by attackers to conceal their activities. The skills and knowledge gained throughout the completion of this assignment will prove valuable to my future professional growth within the cybersecurity industry and, particularly, its spheres of threat detection, incident response, and security monitoring. It is imperative to analyze the logs of the system to detect any signs of breach.

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